3. **Haemagglutinins:** These are proteins in nature and they occur widely in leguminous seeds. Haemagglutinins reduce the food intake resulting in poor growth.
4. **Saponins:** These are present in soyabeans. Saponins cause nausea and vomiting. These toxins can be eliminated by soaking prior to cooking.
5. **Goitrogens:** These substances interfere with iodine uptake by thyroid gland. They are present in soyabean and groundnuts. Excessive intake of these foods may lead to precipitation of goitre.
6. **Tannins:** They have high amount of seed coat in most legumes. Tannins bind with iron irreversibly and interfere with iron absorption. Tannins also bind proteins and reduce their availability.

These toxic constituents can be removed during processing and cooking.

2.10.4. Forms of pulses

Pulses are used in different forms such as:

- Whole legumes
- Decuticled split legumes with and without skin
- Germinated or fermented pulses
- Flour of pulses and
- Parched pulses like Bengal gram and peas.

DO YOU KNOW...?



In December 2013, the United Nations General Assembly (UN) voted unanimously to declare 2016 the International Year of Pulses (IYP). Throughout 2016 the Global Pulse Confederation (GPC) celebrated, raise awareness and promote consumption of pulses at events around the globe.

2.10.5. Soyabean

Soyabean with its high protein contents is considered as a substitute for meat protein which is expensive. Soyabean has 42 grams of protein per 100 grams of the product.

Soyabean can be processed to obtain the following products:

1. **Soya flour:** Soya flour is used in combination with wheat flour in preparation of chapathis. It can also be incorporated in the batter used in the preparation of bajji, vadai and pakoda.



Fig 2.11: Soya

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